

Tarea No. 2.2

2.77 Rebasado

$$V_d = 20 \text{ m/s}$$

$$a = 0.600 \text{ m/s}^2$$

$$t = 45 \text{ min} = 2,700 \text{ s}$$

$$V_c = 20 \text{ m/s}$$

$$t = ?$$

$$x = ?$$

$$V_f = ?$$

$$X_a = 24 \text{ m} + 21 \text{ m} + 26 \text{ m} = 71 \text{ m} + dc$$

$$V_c = 71 \text{ m} + V_c t$$

$$71 \text{ m} + V_c t = \frac{V_d t + at^2}{2}$$

$$X_a = \frac{V_d t + at^2}{2}$$

$$\frac{at^2}{2} = 71 \text{ m}$$

$$t = \sqrt{\frac{2 \cdot 71 \text{ m}}{0.600 \text{ m/s}^2}} = 15.9$$

a) 15.9 s

b) $X_a = 71 \text{ m} + 20 \text{ m/s} \cdot 15.9 \text{ s}$

$X_a = 389 \text{ m}$

c) $V_f = V_d + at$

$$V_f = 20 \text{ m/s} + 0.600 \text{ m/s}^2 \cdot 15.9 \text{ s}$$

$V_f = 29.5 \text{ m/s}$

2.81

$$h_{\max \text{ tierra}} = \frac{V_0^2}{2gt}$$

$$V_0 = \sqrt{-2gt h_{\max t}} \\ V_0 = \sqrt{-2 \cdot 9.8 \text{ m/s}^2 \cdot h} = \sqrt{79.6h}$$

$$h_{\max \text{ marte}} = \frac{-\sqrt{79.6h}}{-3.77 \text{ m/s}^2} = \frac{79.6h}{7.42} = 2.64h$$

$$t_{\text{tierra}} = t$$

$$t_V = t = \frac{-V_0}{g_{\text{tierra}}}$$

$$V_0 = -(9.8 \text{ m/s}^2) t$$

$$t_{\text{marte}} = \frac{-V_0}{g_{\text{marte}}}$$

$$t_{\text{marte}} = \frac{-(9.8 \text{ m/s}^2 t)}{-3.77 \text{ m/s}^2} = 2.64T$$

2.83

$$a) d = \frac{gt^2}{2} = t^2 = \frac{2d}{g}$$

$$t^2 = \frac{2 \cdot 21.3 \text{ m}}{9.8 \text{ m/s}^2} = 4.3 \text{ s}^2 \quad t = \sqrt{4.3} = 2.08 \text{ s}$$

Waar correcte

$$b) t = \frac{V_f}{g} = \frac{2.5 \text{ m/s}}{9.8 \text{ m/s}^2} = 2.55 \text{ s}$$

$$d = \frac{gt^2}{2} = \frac{9.8 \text{ m/s}^2 (2.55 \text{ s})^2}{2} = 31.9 \text{ m}$$

$$\text{Tiempo que elevare} = 31.9 \text{ m} - 21.3 = 10.6 \text{ m}$$

$$V_f^2 - V_0^2 = -2gd$$

$$V_0^2 = 2gd = 2 \cdot 9.8 \text{ m/s}^2 \cdot 10.6 = 207.7 \text{ m}^2/\text{s}^2$$

$$V_0 = 14.4 \text{ m/s}, \text{ no se puede alargar}$$

2.85 ¡Levídalo abajo!

$$V = at \quad y = \frac{at^2}{2}$$

$$y = a \left(\frac{V}{a} \right)^2$$

$$V = \sqrt{2ay} = \sqrt{(2 \cdot 35 \text{ m/s}^2) 0.69 \text{ m}}$$

$$a) V = 6.69 \text{ m/s}$$

$$b) V_f = V_0 - gt \quad h = h_0 + V_0 t - \frac{gt^2}{2}$$

$$t = \frac{V_0}{g} \quad h = \frac{2.20 \text{ m} + (6.69 \text{ m/s}^2)}{12.89 \text{ m/s}^2} = 4.49 \text{ m}$$

$$c) 1.83m = 2.20m + 0.69m/s t - 9.8m/s^2 t^2 / 2$$

$$4.9t^2 - 0.69t - 0.37 = 0, \text{cuadrática}$$

$$t = 1.42 \text{ seg}$$

2.87 Malabarismo

$$a) v_0^2 = v_a^2 + 2g\Delta x$$

$$v_0 = \sqrt{2g h_{\max}} = \sqrt{2 \cdot 9.8m/s^2 \cdot 3} = 7.7m/s$$

$$b) t_{\max} = v_0 / g$$

$$t_{\max} = \frac{7.7m/s}{9.8m/s^2} = 0.78$$

$$c) d = \frac{g}{2} t^2$$

$$3m - d = 5.1m/s t + 4.9m/s^2 t^2$$

$$-3m + 5.1m/s t = 0$$

$$t = 0.59s$$

$$d) d = 4.9m/s$$

$$5g^2 = 1.7m$$

$$3m - 1.7 = 1.3m$$

2.91 Lata gul ul

a) $h = 75 \text{ m}$ $V_f = g t$
 $t = 3.25$

$$V_f = 9.8 \text{ m/s}^2 (3.25 \text{ s}) = \underline{31.85 \text{ m/s}}$$

b) di